Transformation of Variables

1 Model Inadequacies and Transformations

Correcting Model Inadequacies

- Least-squares inference assumes a linear relationship, constant error variance, and normally distributed errors
- Residual plots reveal when these assumptions fail
- One standard remedy is to transform the data; weighting is a separate topic
- In this lecture: transforming y to stabilize the variance, transforming y or x to linearize the model, and two analytical methods for choosing the power (Box–Cox for y, Box–Tidwell for x)

Recognizing Nonconstant Variance

- Plot R-student against the fitted values $\widehat{y}_i$
- Outward-opening funnel: the spread grows with $\mathrm{E}[y_i]$, common when y is a count or an amount
- Double bow: the spread is largest in the middle and small at the extremes, typical when y is a proportion
- With nonconstant variance the OLS estimators remain unbiased but are no longer minimum-variance, and the usual t and F tests are unreliable

Idea of a Variance-Stabilizing Transformation

- Suppose $\mathrm{Var}(y_i)$ depends on the mean $\mathrm{E}[y_i]$
- Replace y by $y^* = h(y)$, with h chosen so that $\mathrm{Var}(y^*)$ is approximately constant
- Then fit the model on the transformed scale

$$y_i^* = \beta_0 + \beta_1 x_i + \varepsilon_i, \qquad \mathrm{Var}(\varepsilon_i) \approx \sigma^2$$

- The appropriate h depends on how $\mathrm{Var}(y)$ is related to $\mathrm{E}[y]$

2 Variance-Stabilizing Transformations

Variance-Stabilizing Transformations

Relationship of σ^2 to $\mathrm{E}[y]$	Transformation
$\sigma^2 \propto$ constant	$y^* = y$ (no transformation)
$\sigma^2 \propto \mathrm{E}[y]$	$y^* = \sqrt{y}$ (Poisson data)
$\sigma^2 \propto \mathrm{E}[y]\,(1 - \mathrm{E}[y])$	$y^* = \sin^{-1}(\sqrt{y}\,)$ (binomial proportions, $0 \le y_i \le 1$)
$\sigma^2 \propto (\mathrm{E}[y])^2$	$y^* = \ln y$
$\sigma^2 \propto (\mathrm{E}[y])^3$	$y^* = y^{-1/2}$
$\sigma^2 \propto (\mathrm{E}[y])^4$	$y^* = y^{-1}$

Reading the Table

- The stronger the dependence of σ^2 on $\mathrm{E}[y]$, the stronger the transformation: $\sqrt{y}$, then $\ln y$, then $y^{-1/2}$, then y^{-1}
- Poisson counts have $\mathrm{Var}(y) = \mathrm{E}[y]$, so $\sigma^2 \propto \mathrm{E}[y]$ and the square root applies
- A proportion $y = X/m$ with $X \sim \mathrm{Binomial}(m, p)$ has $\mathrm{Var}(y) = p(1-p)/m$, so $\sigma^2 \propto \mathrm{E}[y](1 - \mathrm{E}[y])$ and the arcsine square root applies
- $\sigma^2 \propto (\mathrm{E}[y])^2$ means the standard deviation is proportional to the mean, which calls for the logarithm

Choosing a Transformation in Practice

1. Fit the model by OLS on the original y and plot R-student against $\hat{y}_i$
2. If the pattern of spread matches a row of the table, apply that transformation
3. Refit with y^* and examine the residual plots again
4. If the variance is still not constant, try a stronger or weaker transformation

The exact relationship between σ^2 and $\mathrm{E}[y]$ is rarely known, so the choice is often made by trial and error.

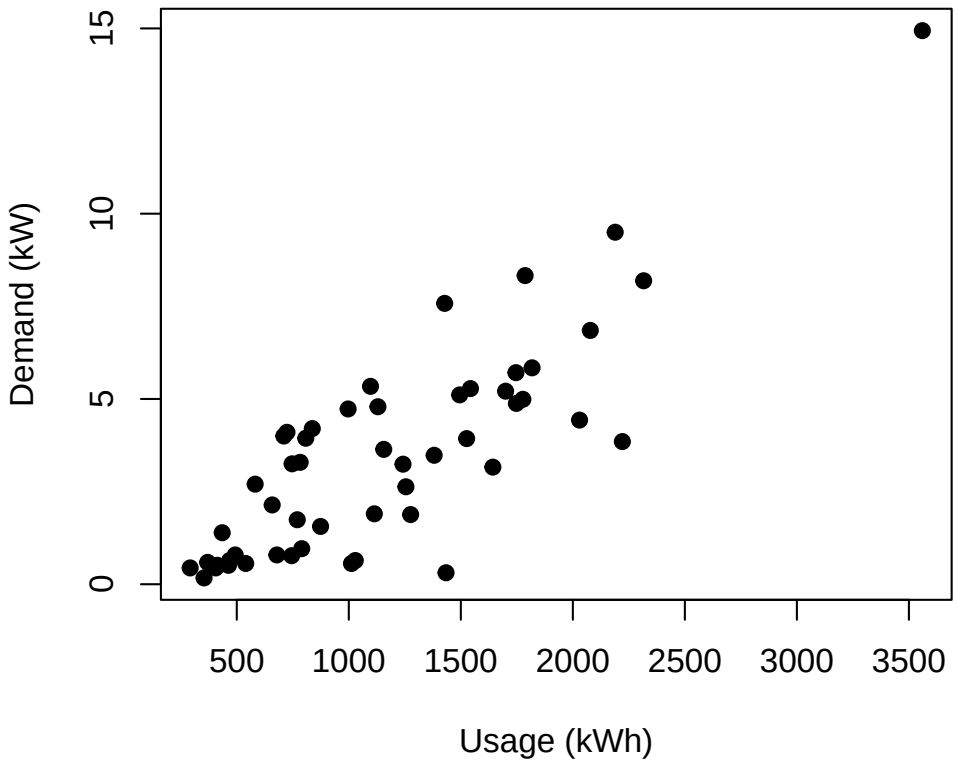
Consequences of Transforming the Response

- Estimates and tests refer to the y^* scale; for example, coefficients describe the change in $\sqrt{y}$, not in y
- To predict on the original scale, apply the inverse transformation to fitted values and to interval endpoints
- R^2 values from models with different responses cannot be compared
- Transforming y also changes the shape of the relationship and the error distribution, so all residual checks must be repeated

3 Example: Electric Utility Data

Electric Utility Data

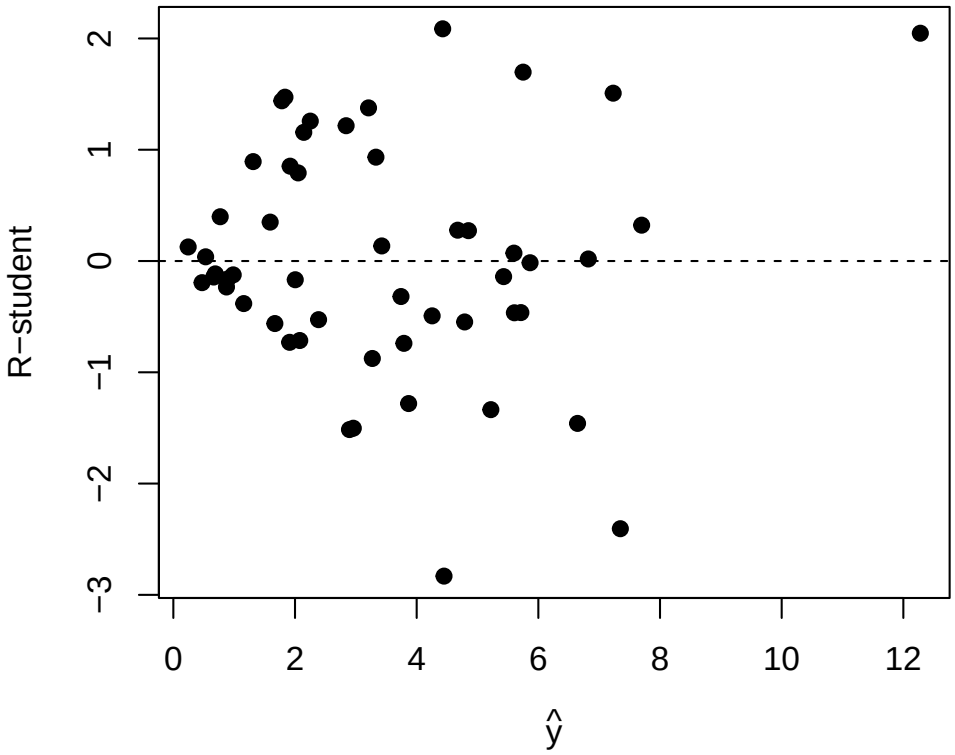
- y: peak demand (kW), x: monthly energy usage (kWh)
- $n = 53$ residential customers in August (MPV Table 5.2)
- Goal: model demand as a function of usage
- The spread of y increases with x



Straight-Line Fit

$$\hat{y} = -0.8313 + 0.003683\,x$$
$$R^2 = 0.7046$$

- R-student against $\hat{y}$ shows an outward-opening funnel
- The constant-variance assumption is violated



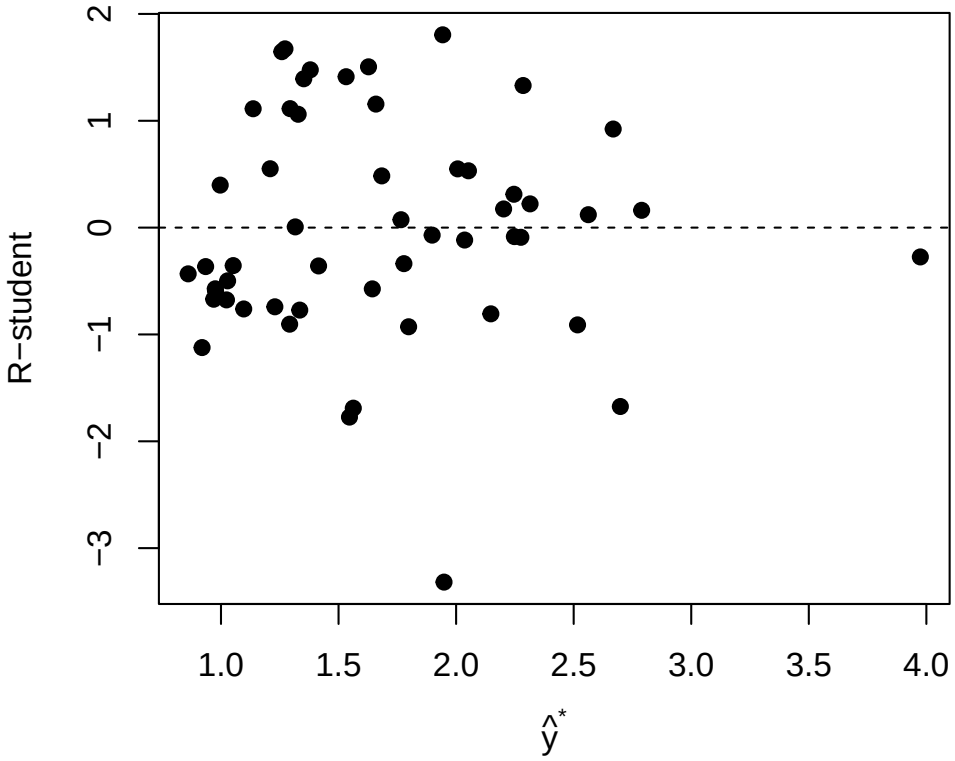
Question: Which Transformation?

The residual plot is an outward-opening funnel, so the variance increases with the mean. Which row of the variance-stabilizing table would you try first, and what would you do if the funnel persisted?

Square-Root Transformation

$$
\begin{aligned}
y^* &= \sqrt{y} \\
\hat{y}^* &= 0.5822 + 0.0009529\, x \\
R^2 &= 0.6485
\end{aligned}
$$

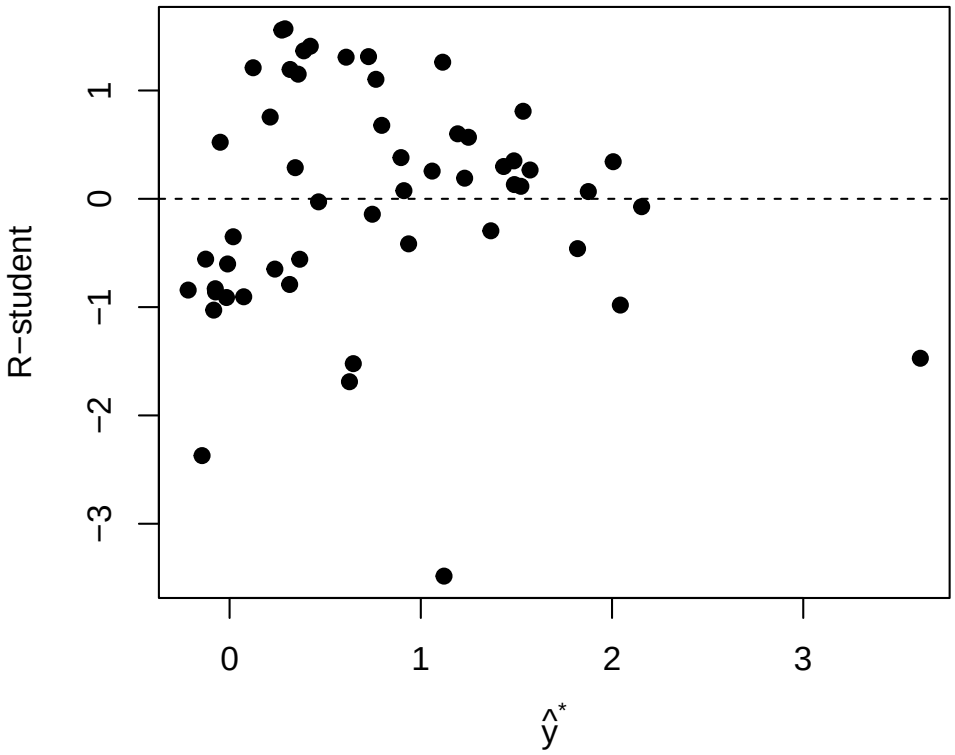
- R-student against $\hat{y}^*$ no longer shows a funnel
- The variance now looks stable



A Stronger Transformation: $\ln y$

$$
\begin{aligned}
y^* &= \ln y \\
\hat{y}^* &= -0.5587 + 0.001172\, x \\
R^2 &= 0.5266
\end{aligned}
$$

- The residual plot now shows curvature
- The logarithm is too strong for these data



Question: Did R^2 Get Worse?

R^2 dropped from 0.7046 for y to 0.6485 for $\sqrt{y}$. Does this mean the transformed model is worse?

4 Transformations to Linearize the Model

Linearizable Nonlinear Functions

Function	Transformation	Linear form
$y = \beta_0 x^{\beta_1}$	$y' = \ln y,\ x' = \ln x$	$y' = \ln \beta_0 + \beta_1 x'$
$y = \beta_0 e^{\beta_1 x}$	$y' = \ln y$	$y' = \ln \beta_0 + \beta_1 x$
$y = \beta_0 + \beta_1 \ln x$	$x' = \ln x$	$y = \beta_0 + \beta_1 x'$
$y = \dfrac{x}{\beta_0 x - \beta_1}$	$y' = \dfrac{1}{y},\ x' = \dfrac{1}{x}$	$y' = \beta_0 - \beta_1 x'$

The Error Structure Matters

- Linearizing the exponential function assumes the error is multiplicative

$$y = \beta_0 e^{\beta_1 x}\,\varepsilon \quad \Longrightarrow \quad \ln y = \ln \beta_0 + \beta_1 x + \ln \varepsilon$$

- The linearized model is appropriate if $\ln \varepsilon$ has mean zero, constant variance, and is normal
- If instead the error is additive, $y = \beta_0 e^{\beta_1 x} + \varepsilon$, taking logarithms does not give a linear model with additive error, and nonlinear regression is needed

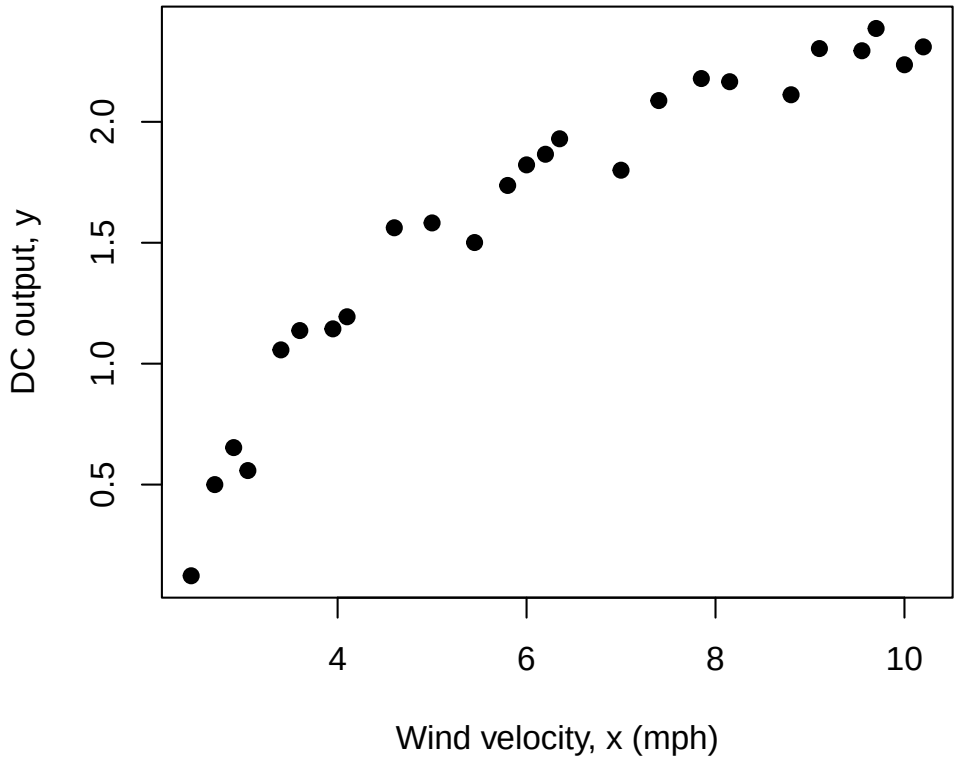
Transform x or Transform y?

- Transforming x alone leaves the distribution of the errors unchanged
- Transforming y changes both the shape of the relationship and the error distribution
- Curved scatter with roughly constant variance: transform x
- Curved scatter with nonconstant variance: transform y, which may fix both problems

5 Example: Windmill Data

Windmill Data

- y: DC output of a windmill, x: wind velocity (mph)
- $n = 25$ observations (MPV Table 5.5)
- Goal: model DC output as a function of wind velocity
- The scatter plot is clearly curved



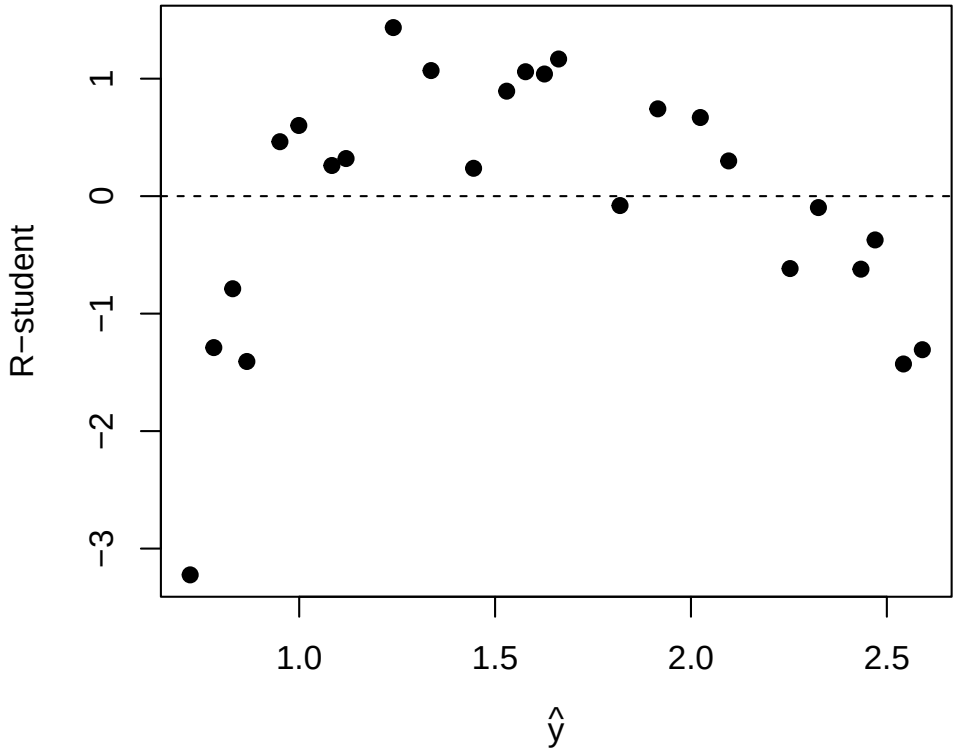
Question: Which Transformation?

DC output rises with wind velocity but levels off, and the spread around the trend looks roughly constant. Which transformation would you try, and on which variable?

Straight-Line Fit

$$\hat{y} = 0.1309 + 0.2411\,x$$
$$R^2 = 0.8745$$

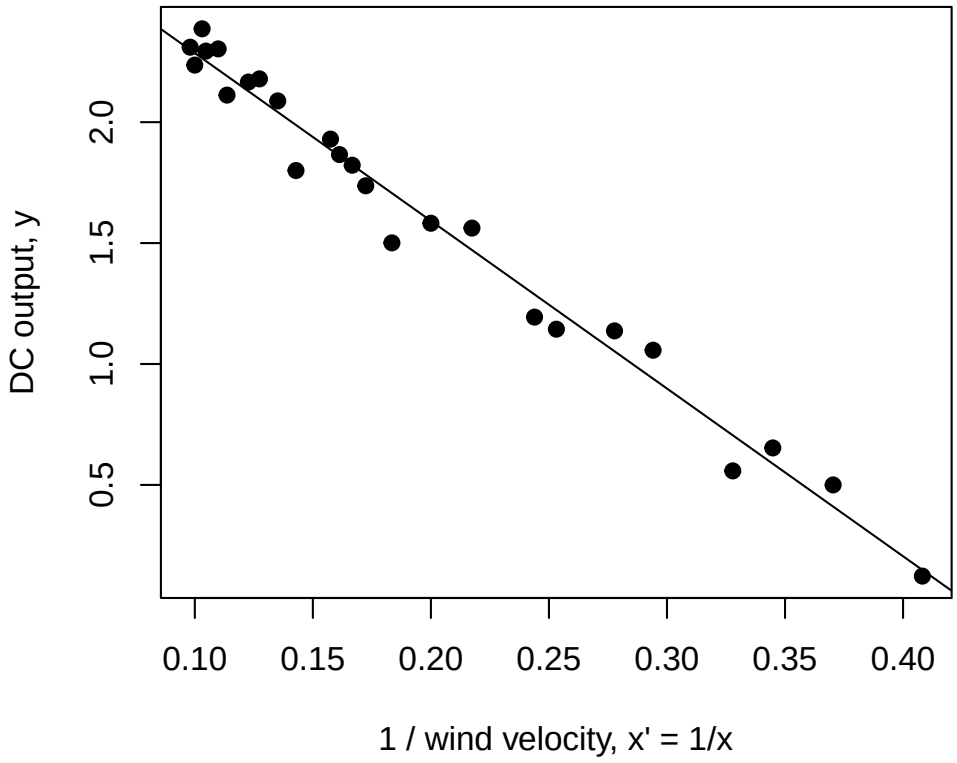
- R-student against $\hat{y}$ shows a clear curved pattern
- The straight line is inadequate despite the high R^2



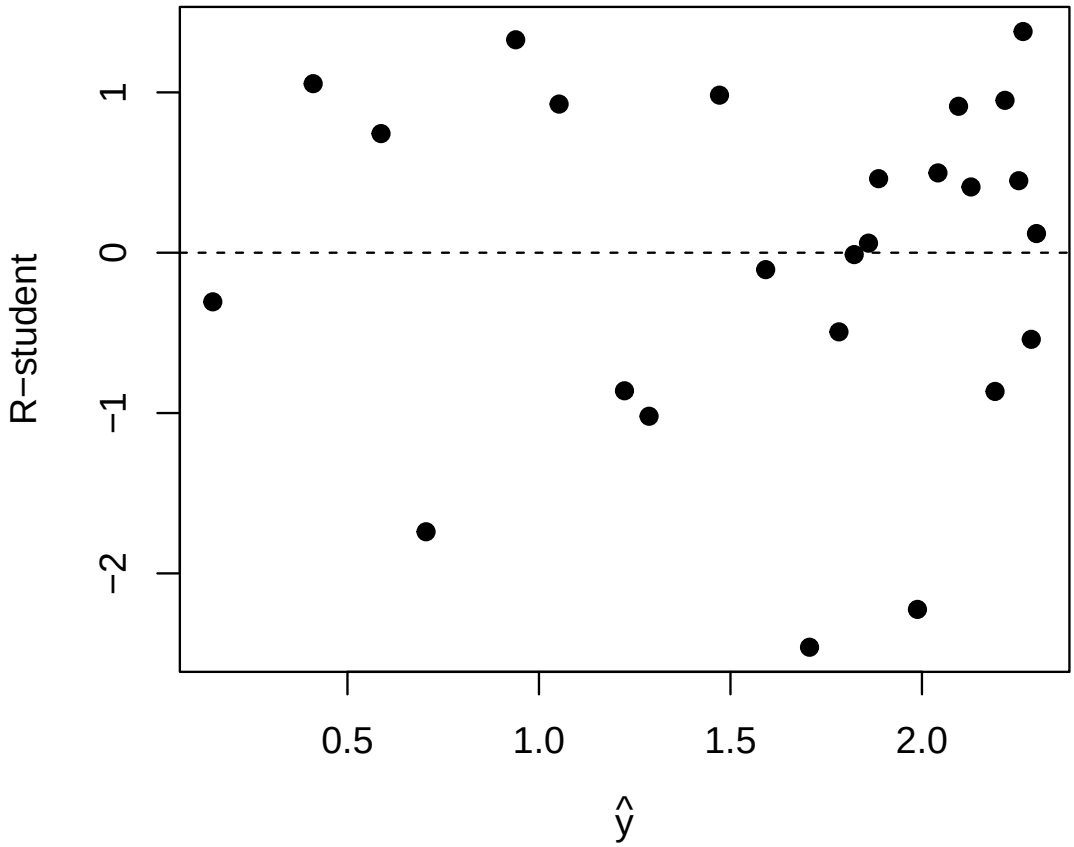
Reciprocal Transformation of x

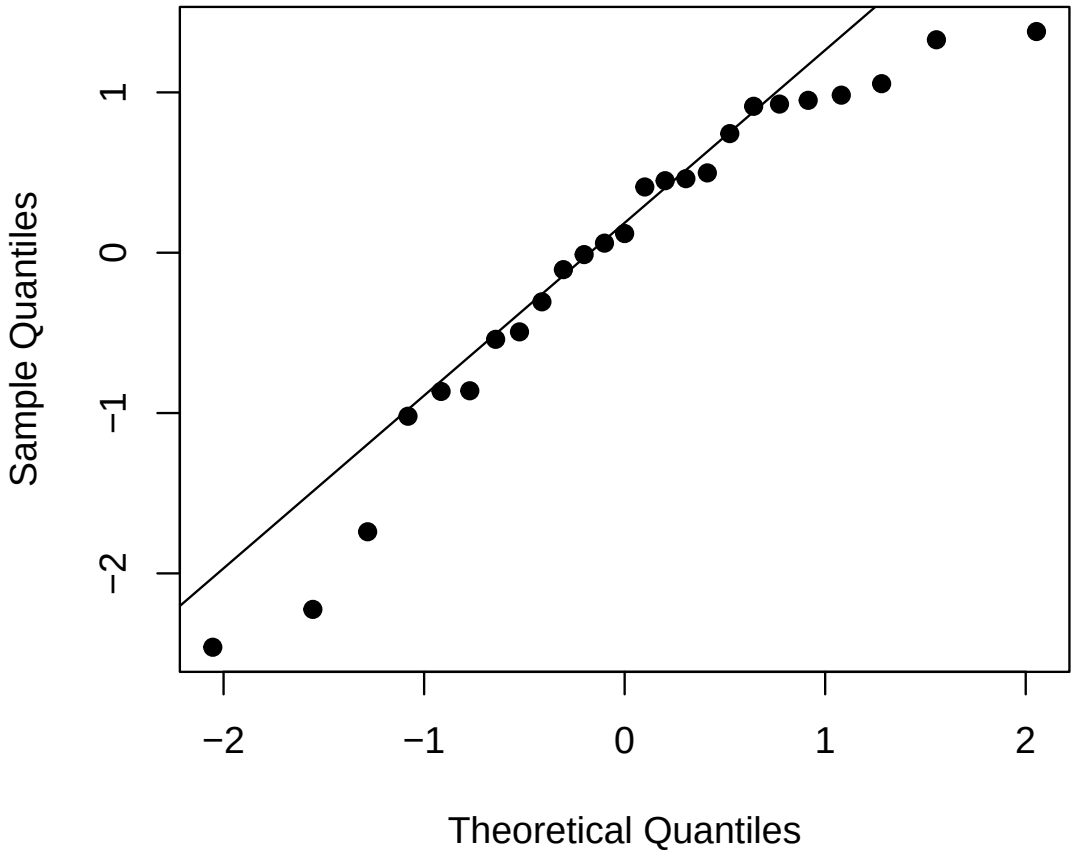
$$x' = 1/x$$
$$\hat{y} = 2.9789 - 6.9345\,x'$$
$$R^2 = 0.9800$$

- The scatter plot against x' is close to a straight line
- As $x \to \infty$ the fitted output approaches 2.9789, the upper limit



Residual Analysis After the Transformation


R−student
1
0
−1
−2
0.5
1.0
1.5
2.0
ŷ



- R-student against $\hat{y}$ shows no systematic pattern
- The normal probability plot is close to a straight line

6 The Box–Cox Method

Motivation for the Box–Cox Method

- The variance-stabilizing table requires knowing how σ^2 depends on $\mathrm{E}[y]$, which is usually unknown
- Box and Cox (1964): estimate the power of the transformation from the data
- Considers the power family y^λ, with λ estimated together with $\boldsymbol{\beta}$ by maximum likelihood
- Aims to correct nonnormality and nonconstant variance, and requires $y_i > 0$

The Power Transformation Family

$$y^{(\lambda)} = \begin{cases} \dfrac{y^\lambda - 1}{\lambda}, & \lambda \neq 0 \\ \ln y, & \lambda = 0 \end{cases}$$

- As $\lambda \to 0$, $(y^\lambda - 1)/\lambda \to \ln y$, so the logarithm belongs to the family
- $\lambda = 1$: no transformation, $\lambda = 1/2$: square root, $\lambda = -1$: reciprocal

The Scaled Response

- The size of y^λ changes with λ, so $\mathrm{SS}_{\mathrm{Res}}$ is not comparable across values of λ
- Fix: divide by a factor involving the geometric mean $\dot{y}$

$$y^{(\lambda)} = \begin{cases} \dfrac{y^\lambda - 1}{\lambda\, \dot{y}^{\lambda-1}}, & \lambda \neq 0 \\ \dot{y} \ln y, & \lambda = 0 \end{cases} \qquad \dot{y} = \exp\left[\frac{1}{n}\sum_{i=1}^{n} \ln y_i\right]$$

Model and Estimation

$$\mathbf{y}^{(\lambda)} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \qquad \boldsymbol{\varepsilon} \sim N(\mathbf{0}, \sigma^2\mathbf{I})$$

- For fixed λ, $\widehat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}^{(\lambda)}$
- Let $\mathrm{SS}_{\mathrm{Res}}(\lambda)$ be the residual sum of squares from this fit

$$\mathrm{SS}_{\mathrm{Res}}(\lambda) = \sum_{i=1}^{n} \left(y_i^{(\lambda)} - \widehat{y}_i^{(\lambda)}\right)^2$$

- With the scaled response, the maximum likelihood estimate $\hat{\lambda}$ is the value that minimizes $\mathrm{SS}_{\mathrm{Res}}(\lambda)$

The Box–Cox Procedure

1. Choose a grid of values of λ, for example from -2 to 2
2. For each λ, compute the scaled $y^{(\lambda)}$, fit the model, and record $\mathrm{SS}_{\mathrm{Res}}(\lambda)$
3. Plot $\mathrm{SS}_{\mathrm{Res}}(\lambda)$ against λ and locate the minimum $\hat{\lambda}$
4. Analyze the simple power y^λ (not the scaled version), using a convenient value of λ near $\hat{\lambda}$, and recheck the residuals

Later inference treats λ as known, so it is only approximate.

Confidence Interval for λ

$$\mathrm{SS}^* = \mathrm{SS}_{\mathrm{Res}}(\hat{\lambda})\left(1 + \frac{t^2_{\alpha/2,\,\nu}}{\nu}\right)$$

- ν is the residual degrees of freedom
- A $100(1-\alpha)\%$ confidence interval for λ contains all λ with $\mathrm{SS}_{\mathrm{Res}}(\lambda) \le \mathrm{SS}^*$
- If the interval contains 1, no transformation is needed
- Otherwise choose a convenient value inside the interval, such as -1, 0, $1/2$

Log-Likelihood Form and `MASS::boxcox`

- Up to a constant, the profile log-likelihood of λ is

$$L(\lambda) = -\frac{n}{2} \ln \mathrm{SS}_{\mathrm{Res}}(\lambda)$$

- The likelihood-based interval keeps all λ with $L(\hat{\lambda}) - L(\lambda) \leq \frac{1}{2}\chi^2_{\alpha,\,1}$
- Equivalently, $\mathrm{SS}_{\mathrm{Res}}(\lambda) \leq \mathrm{SS}_{\mathrm{Res}}(\hat{\lambda}) \exp(\chi^2_{\alpha,\,1}/n)$, which agrees closely with SS^* unless n is small
- `MASS::boxcox` plots $L(\lambda)$ with this 95% interval

7 Example: Box–Cox on the Electric Utility Data

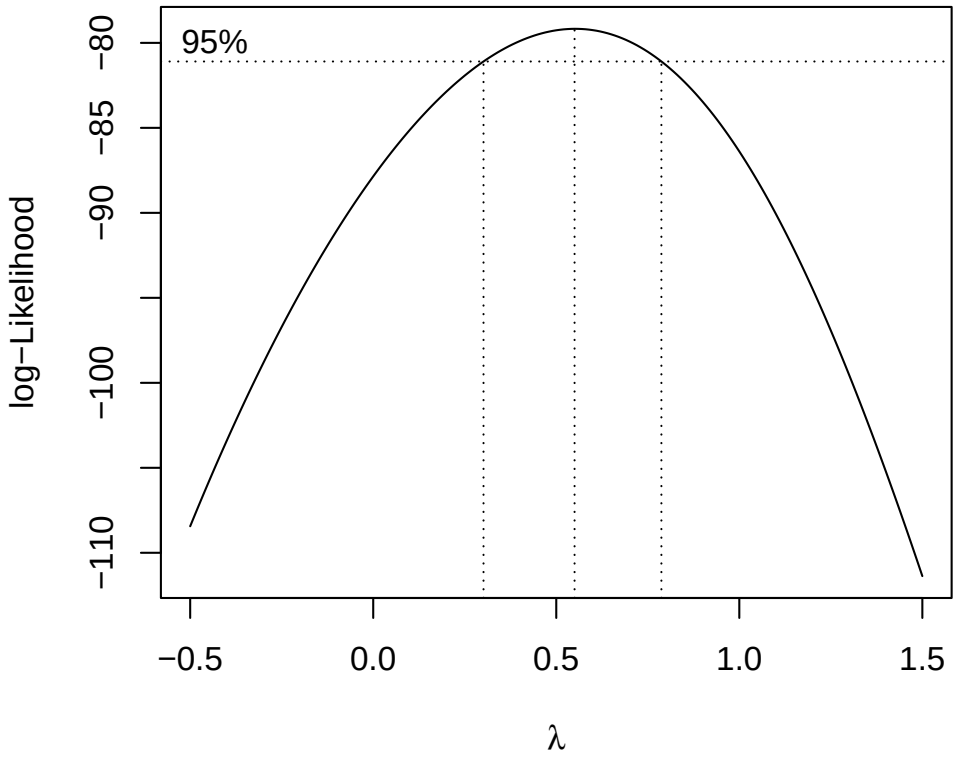
$\mathrm{SS}_{\mathrm{Res}}(\lambda)$ for the Electric Utility Data

λ	$\mathrm{SS}_{\mathrm{Res}}(\lambda)$
-1	986.03
-0.5	291.58
0	134.09
0.25	107.21
0.5	96.95
0.75	101.69
1	126.87

- The scaled $\mathrm{SS}_{\mathrm{Res}}(\lambda)$ is smallest near $\lambda = 0.5$

Box–Cox Plot from `MASS`

- $\hat{\lambda} = 0.55$
- 95% interval: $(0.31,\ 0.78)$
- Contains $\lambda = 0.5$
- Excludes $\lambda = 1$ and $\lambda = 0$



Question: Which λ Should We Use?

The estimate is $\widehat{\lambda} = 0.55$ with 95% interval $(0.31, 0.78)$. Which value would you use for the final model, and why?

- Use $\lambda = 0.5$, that is $\sqrt{y}$: it lies inside the interval and is easy to interpret
- It agrees with the transformation chosen from the residual plots and the variance-stabilizing table
- The logarithm ($\lambda = 0$) and no transformation ($\lambda = 1$) are both outside the interval

8 The Box–Tidwell Method

Motivation for the Box–Tidwell Method

- Box–Cox transforms y; when the problem is nonlinearity with constant variance, transform the regressor instead
- Model, with unknown power α:

$$y = \beta_0 + \beta_1 \xi + \varepsilon, \qquad \xi = \begin{cases} x^\alpha, & \alpha \neq 0 \\ \ln x, & \alpha = 0 \end{cases}$$

- Box and Tidwell (1962) estimate α iteratively
- Special case: $\alpha = 0$: logarithm

Linearizing the Model in α: Setup

- The model $\mathrm{E}[y] = \beta_0 + \beta_1 x^\alpha$ is nonlinear in the unknown power α
- Idea: replace it by a first-order Taylor series in α about a starting value α_0, treating x, β_0, β_1 as fixed

$$f(\alpha) \approx f(\alpha_0) + (\alpha - \alpha_0)\, f'(\alpha_0), \qquad f(\alpha) = \beta_0 + \beta_1 x^\alpha$$

- Take $\alpha_0 = 1$: then $x^{\alpha_0} = x$, the untransformed straight-line model
- We need $f'(\alpha)$, which requires $\partial x^\alpha / \partial \alpha$

Step 1: Differentiate x^α with Respect to α

- Here x is a constant and α is the variable, so x^α is an exponential function of α, not a power function of x
- Rewrite it with base e (this requires $x > 0$)

$$x^\alpha = e^{\alpha \ln x}$$

- $\ln x$ is a constant with respect to α, so the chain rule gives

$$\frac{\partial}{\partial \alpha}\, x^\alpha = \frac{\partial}{\partial \alpha}\, e^{\alpha \ln x} = (\ln x)\, e^{\alpha \ln x} = x^\alpha \ln x$$

- Do not confuse this with $\partial x^\alpha / \partial x = \alpha x^{\alpha - 1}$, which differentiates with respect to x

Step 2: Evaluate the Taylor Expansion at $\alpha_0 = 1$

β_0 and β_1 do not depend on α, so differentiating $f(\alpha) = \beta_0 + \beta_1 x^\alpha$ gives

$$f'(\alpha) = \beta_1 \frac{\partial x^\alpha}{\partial \alpha} = \beta_1\, x^\alpha \ln x$$

Evaluate f and f' at $\alpha_0 = 1$:

$$f(1) = \beta_0 + \beta_1 x, \qquad f'(1) = \beta_1\, x \ln x$$

Substitute into $f(\alpha) \approx f(1) + (\alpha - 1)\, f'(1)$:

$$\mathrm{E}[y] \approx \beta_0 + \beta_1 x + (\alpha - 1)\, \beta_1\, x \ln x$$

Step 3: Reparameterize to a Linear Model

- $(\alpha - 1)\beta_1$ is a single unknown constant, and $x \ln x$ can be computed from the data

$$\gamma = (\alpha - 1)\beta_1, \qquad w = x \ln x$$

$$\mathrm{E}[y] \approx \beta_0 + \beta_1 x + \gamma w$$

- The model is linear in the regressors x and w, so the coefficients can be estimated by OLS
- Recover the power by solving $\gamma = (\alpha - 1)\beta_1$ for α

$$\alpha = \frac{\gamma}{\beta_1} + 1$$

- $\gamma = 0$ exactly when $\alpha = 1$, that is, when no transformation of x is needed

One Iteration of the Procedure

1. Fit $y = \beta_0 + \beta_1 x + \varepsilon$ and record $\hat{\beta}_1$
2. Fit $y = \beta_0^* + \beta_1^* x + \gamma w + \varepsilon$ with $w = x \ln x$ and record $\hat{\gamma}$
3. Compute the first estimate of the power

$$\hat{\alpha}_1 = \frac{\hat{\gamma}}{\hat{\beta}_1} + 1$$

Iterating

- Set $x^{(1)} = x^{\hat{\alpha}_1}$ and repeat the procedure with $x^{(1)}$ in place of x
- Stop when $\hat{\alpha}_k \approx 1$, that is, when no further transformation is needed
- The overall estimate is the product $\hat{\alpha} = \prod_k \hat{\alpha}_k$, since a power of a power multiplies the exponents
- The t statistic for $\hat{\gamma}$ gives an approximate test of $H_0 \colon \alpha = 1$

9 Example: Box–Tidwell on the Windmill Data

Windmill Data: First Iteration

$$\hat{\beta}_1 = 0.2411, \qquad \hat{\gamma} = -0.4626 \quad (t = -9.13)$$

$$\hat{\alpha}_1 = \frac{-0.4626}{0.2411} + 1 = -0.918$$

- The significant $\hat{\gamma}$ confirms that the straight line is inadequate
- $\hat{\alpha}_1$ is close to -1, which suggests the reciprocal transformation

Windmill Data: Iterations

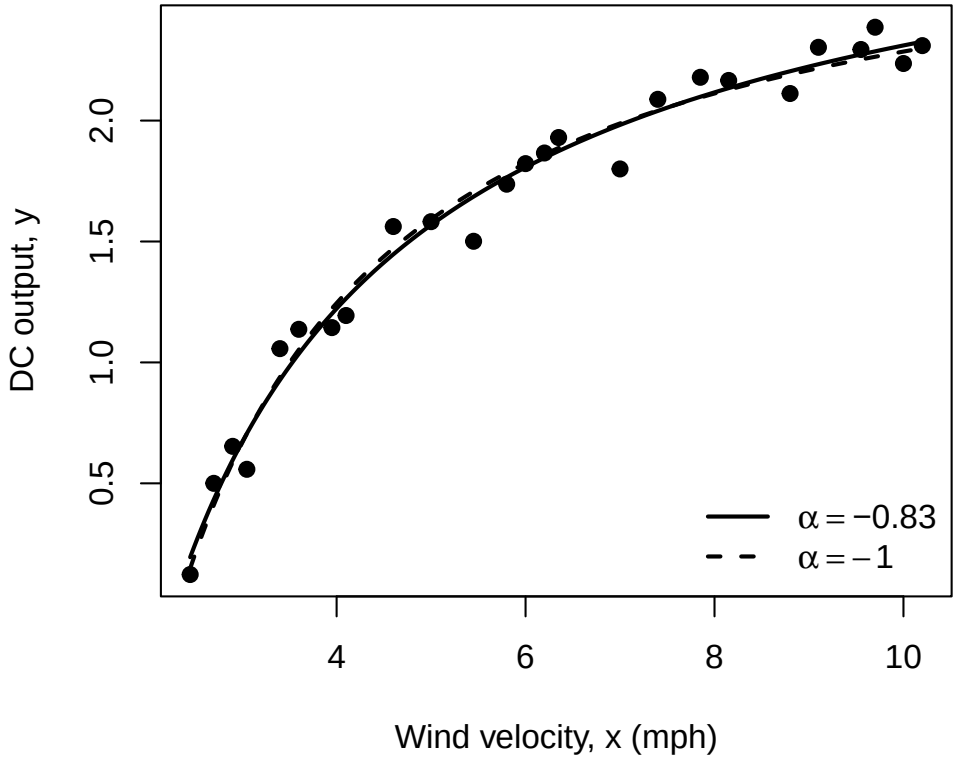
k	$\hat{\beta}_1$	$\hat{\gamma}$	$t(\hat{\gamma})$	$\hat{\alpha}_k$	cumulative $\hat{\alpha}$
1	0.2411	−0.4626	−9.13	−0.9183	−0.9183
2	−6.6784	0.5994	0.51	0.9103	−0.8359
3	−6.4738	0.0190	0.02	0.9971	−0.8334
4	−6.4686	0.0006	0.00	0.9999	−0.8333

- From the second iteration on, $\hat{\gamma}$ is not significant and $\hat{\alpha}_k \approx 1$
- The procedure converges to $\hat{\alpha} = -0.833$

Box–Tidwell Fit Compared with the Reciprocal Model

$$\hat{y} = 3.2601 - 6.4684\, x^{-0.833}$$

	$\alpha = -0.833$	$\alpha = -1$
R^2	0.9809	0.9800
$\hat{\sigma}$	0.0920	0.0942



Question: $x^{-0.833}$ **or** $1/x$**?**

The estimated power is -0.833, but the reciprocal model has almost the same R^2. Which model would you report?

- The reciprocal model: R^2 and $\hat{\sigma}$ are essentially unchanged and the fitted curves nearly coincide
- $1/x$ is simpler to interpret and the intercept is the upper limit of DC output

10 Summary

Summary

- Nonconstant variance: transform y using the variance-stabilizing table, then recheck the residuals
- Unknown variance–mean relationship: estimate the power of y with the Box–Cox method

- Curved relationship with constant variance: transform x, guided by the shape of the scatter plot
- Unknown power for x: estimate it with the Box–Tidwell method
- After any transformation, repeat the residual analysis and interpret results on the transformed scale